

6 Designing the circuit

Now that we have decided that we will be using a pic microcontroller and two IR sensors, in this section we will understand the circuit.

Circuit for the sensor

The IR sensor will require one IR transmitter and one IR receiver. The IR transmitter will target its radiations on the surface below it. When there is white surface below the sensor, it would reflect most of the radiations incident on it (because white colour is almost a perfect reflector). Some of these radiations would reach the IR receiver and the IR receiver would then start conducting. In that case, the terminal two of the Opamp would be grounded. Terminal three is grounded throughout. So, the potential difference between the terminal two and terminal three would be very near to zero. The Opamp output terminal would thus give zero voltage as its output. This output is logic zero. But when the surface below the sensor is black, the radiations emitted by IR transmitter would not be reflected back by the black surface (black colour is almost a perfect absorber). So, no radiations are received by IR receiver and it offers open circuit. In that case, terminal two will remain connected to high voltage, and the difference between terminal two and terminal three (which is grounded) would be maximum (nearly 3 Volts). This difference is amplified by the Opamp and the output would then give around 5 volt at terminal one. This output is logic one.

Note that a LED is also used in the circuit for sensor. It is just to check that whether sensor is responding to the black and white surfaces. The LED should not glow when there is white surface detection (logic zero, thus no voltage at output of the Opamp), and it should glow when the sensor detects black surface (logic one, thus 5V at output of Opamp.)

Now, let us study the complete circuit. The pic microcontroller has been provided with a clock of 4 MHz with the help of crystal oscillator. This method is already discussed before.

We are going to use PORT A of PIC 16F84 as

